Heltec WiFi LoRa 32 V4 (plugs into J2/J3)

#X_PACK
Pack board (P5)
XH-6

J_PWR1
PWR

#X_HBAT
Heltec V4 battery
XH-2 to the module's SH1.25

J_HBAT1
HELTEC_BAT

C_BULK1 47uF 10V
C_BULK2 47uF 10V
C_5V1 10uF
C_5V2 100nF

J2
Heltec_J2

J3
Heltec_J3

GPIO expander (0x20), button inputs

Low-side drivers: button LEDs, buzzer

NS4168 supply and shutdown

U3 NS4168

J_LCD1
LCD
#X_LCD
LCD 1602 + PCF8574
XH-4

J_NFC1
NFC
#X_NFC
PN532 (I2C)
XH-4

C_NFC1 100uF 6.3V

J_BTN_R1
BTN_R
#X_BTN_R
ONPOW LAS1-AGQ-11E/R/6V
XH-4

J_BTN_B1
BTN_B
#X_BTN_B
ONPOW LAS1-AGQ-11E/B/6V
XH-4

J_BUZ1
BUZ
#X_BUZ
BZ-38 buzzer
XH-2

J_SPK1
SPK
#X_SPK
VISATON K 50, 8 ohm
XH-2, BTL: neither side is GND

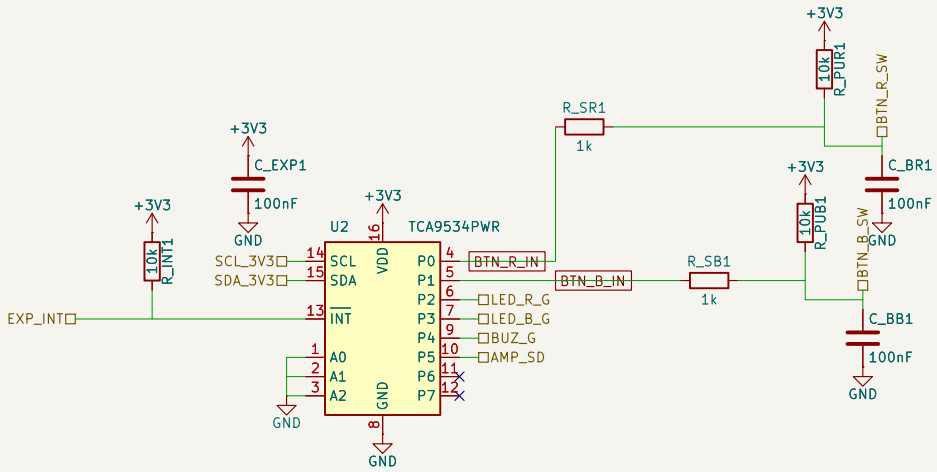
J_UART1
UART

J_I2C1
I2C

H1 M3
H2 M3
H3 M3
H4 M3

| | |
|---|---|
| Sheet: / | |
| File: carrier.kicad_sch | |
| **Title: AirsoftCounter v2 carrier: system** | |
| Size: A3 | Date: |
| KiCad E.D.A. 10.0.7 | Rev: |
| | Id: 1/4 |

## GPIO expander (0x20), button inputs

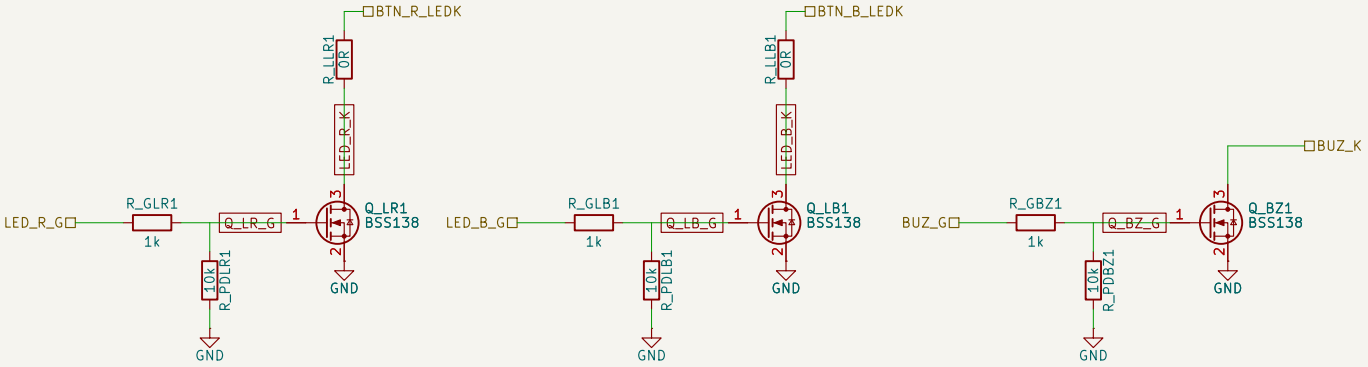



Sheet: /GPIO expander (0x20), button inputs/
File: carrier-expander.kicad_sch

**Title: AirsoftCounter v2 carrier: GPIO expander (0x20), button inputs**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 2/4 |

## Low-side drivers: button LEDs, buzzer

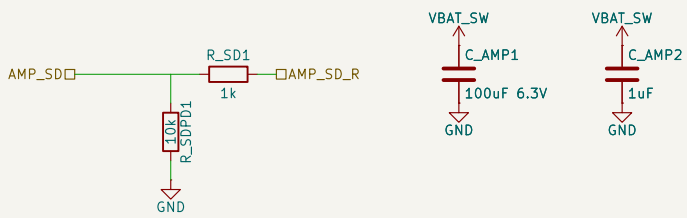



Sheet: /Low-side drivers: button LEDs, buzzer/
File: carrier-drivers.kicad_sch

**Title: AirsoftCounter v2 carrier: Low-side drivers: button LEDs, buzzer**

| Size: A4 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/4 |

## NS4168 supply and shutdown




| | | |
|---|---|---|
| Sheet: /NS4168 supply and shutdown/ | | |
| File: carrier-audio.kicad_sch | | |
| **Title: AirsoftCounter v2 carrier: NS4168 supply and shutdown** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.7 | | Id: 4/4 |